

Math 112 – Calculus I

Section 2.7: Derivatives and Rates of Change

From average change to change at an instant

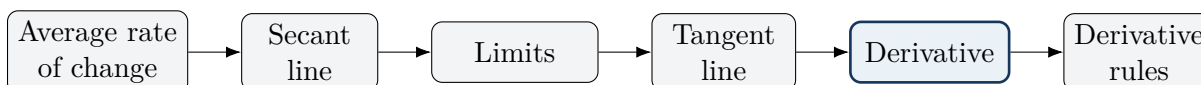
Last Time

Limits help us understand what happens as we approach a point. Today we'll use this idea to answer a new question: how can we measure change at a single instant?

Today

By the end of today's lecture, you should be able to:

- distinguish a secant line from a tangent line;
- explain the derivative as a slope and as an instantaneous rate of change;
- use limits to compute the derivative at a point;
- find the equation of a tangent line;
- interpret derivatives in a motion problem, including correct units.



Think–Pair–Share

Imagine you are driving a car and your speedometer reads 63mph. How does your car know your speed **right now**?

1. Reviewing Lines and Slopes

Think–Pair–Share

What is the difference between a tangent line and a secant line?
How do you find the equation of a tangent line?

Strategy – Equation of a line

Once you know a point (x_0, y_0) and a slope m , use point-slope form: $y - y_0 = m(x - x_0)$.

Example 1: Warm-up with point-slope form

A tangent line passes through $(-3, 4)$ and has slope $\frac{2}{3}$. Write the equation of the tangent line.

We now discuss the question, how do I find the slope of a line at a single point?

Strategy–Slope of a Secant Line

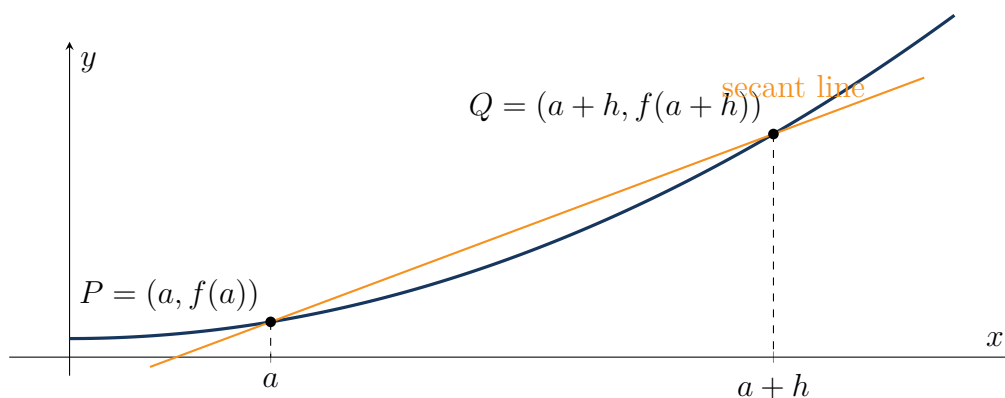
The slope of the secant line between the points (x_0, y_0) and (x, y) is given by

$$m_{\text{sec}} = \frac{y - y_0}{x - x_0}.$$

This represents the **average** rate of change from x_0 to x .

For points $P = (a, f(a))$ and $Q = (a + h, f(a + h))$, the slope of the secant line is

$$m_{\text{sec}} = \frac{f(a + h) - f(a)}{(a + h) - a} = \frac{f(a + h) - f(a)}{h}.$$



As h approaches 0, the point Q moves toward P . The secant lines approach the tangent line.

2. Tangent Lines

Definition – Tangent Line

The **tangent line** to $y = f(x)$ at a point $P = (a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

provided the limit exists. Equivalently,

$$m = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

Note: This is the same equation as above with $h = x - a$.

This is the **instantaneous** rate of change of f at $x = a$.

Common Mistake

The tangent line is *not* simply a line that touches the graph once. The important feature is that its slope is the limiting slope of nearby secant lines.

3. The Derivative at a Point**Definition – Derivative at a**

The derivative of f at a , denoted $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

or equivalently

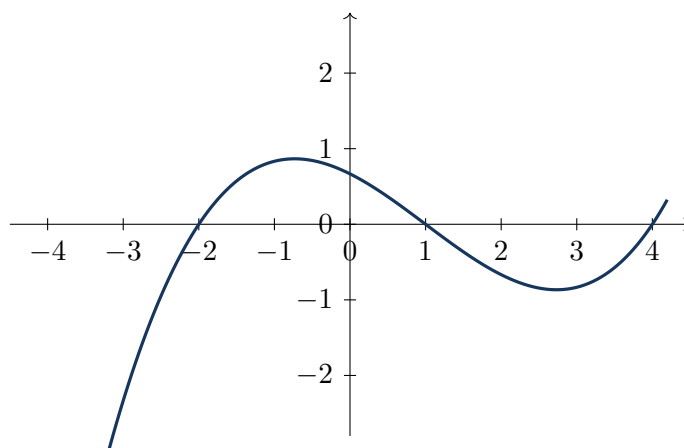
$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

provided the limit exists.

Big Idea

The derivative of f at a ($f'(a)$) can be interpreted as:

- the slope of the tangent line at $x = a$.
- the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

**Think–Pair–Share**

For the function graphed above discuss whether the derivative is positive or negative at each of the following values, and which is largest:

1. $x = -3$
2. $x = 1$
3. $x = 3.5$

Strategy – Finding $f'(a)$ from the definition

1. Write $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.
2. Compute $f(a+h)$ and $f(a)$ carefully.
3. Simplify the difference quotient. Do not substitute $h = 0$ too early.
4. Evaluate the limit.
5. Interpret the result and include units when appropriate.

Example 2A: Calculating Derivatives

Use the limit definition to find the slope of the tangent line to the curve at the given point (a, b) . (This is the same as finding the derivative of the function at $x = a$)

$f(x) = x^2$ at $(2, 4)$

Common Mistake

A common algebra error is to write $(a+h)^2 = a^2 + h^2$. Remember:

$$(a+h)^2 = a^2 + 2ah + h^2.$$

The middle term is what allows the difference quotient to simplify.

Example 2B:

Find the slope of the tangent line at the given point: $y = f(x) = -x^2 + 2x + 2$ at $x = 3$.

Example 2C:

Find the slope of the tangent line at the given point: $f(x) = \frac{1}{x}$ at $(2, \frac{1}{2})$. After finding the slope, also find the equation of the tangent line.

Common Mistake

For a specified $x = a$, the derivative $f'(a)$ is a **number**; it is the slope at one point. Later, $f'(x)$ will denote a new **function** giving the slope at every allowable x .

Example 3: A formula for the derivative at a general point

Let $f(x) = x^2 - 8x + 9$.

Part A. Find $f'(a)$ from the definition. Therefore, $f'(a) = \underline{\hspace{2cm}}$.

Part B. Find the tangent line when $x = 2$.

$$f(2) = \underline{\hspace{2cm}}, \quad f'(2) = \underline{\hspace{2cm}}.$$

Hence,

$\underline{\hspace{4cm}}$.

4. Average and Instantaneous Velocity

Think–Pair–Share

Suppose a 60-mile trip takes exactly one hour.

1. Was the car necessarily traveling at exactly 60 mph throughout the trip?
2. What does the number 60 mph describe?
3. How is average velocity calculated?
4. What does a speedometer describe?

Big Idea

An **average rate of change** compares two different input values. An **instantaneous rate of change** describes how quickly a quantity is changing at one particular input value.

Derivatives can also be thought of as velocity.

Strategy

The derivative of a position function is the instantaneous rate of change of the position, which is the instantaneous velocity (or just velocity for short).

If $s(t)$ gives position, then the average velocity between a and $a + h$ is

$$v_{avg} = \frac{s(a + h) - s(a)}{h}.$$

Taking the limit gives the instantaneous velocity.

Definition – Instantaneous Velocity

For an object with position function $s(t)$, the instantaneous velocity at time $t = a$ is

$$v(a) = \lim_{h \rightarrow 0} v_{avg} = \lim_{h \rightarrow 0} \frac{s(a + h) - s(a)}{h} = s'(a),$$

provided the limit exists.

Guided Investigation: A falling object

Suppose an object falling in a vacuum has position

$$s(t) = 16t^2,$$

where s is measured in feet and t in seconds.

Part A. Compute some positions. How far did the object fall after 1 sec? 2 sec?

$s(1) = \rule{1.5cm}{0.4pt}$ ft, $s(2) = \rule{1.5cm}{0.4pt}$ ft.

Part B. Complete the average-velocity table beginning at $t = 1$.

Interval	Average velocity	Value (ft/s)
$[1, 2]$	$\frac{s(2) - s(1)}{2 - 1}$	$\rule{1.5cm}{0.4pt}$
$[1, 1.5]$	$\frac{s(1.5) - s(1)}{0.5}$	$\rule{1.5cm}{0.4pt}$
$[1, 1.1]$	$\frac{s(1.1) - s(1)}{0.1}$	$\rule{1.5cm}{0.4pt}$
$[1, 1.01]$	$\frac{s(1.01) - s(1)}{0.01}$	$\rule{1.5cm}{0.4pt}$

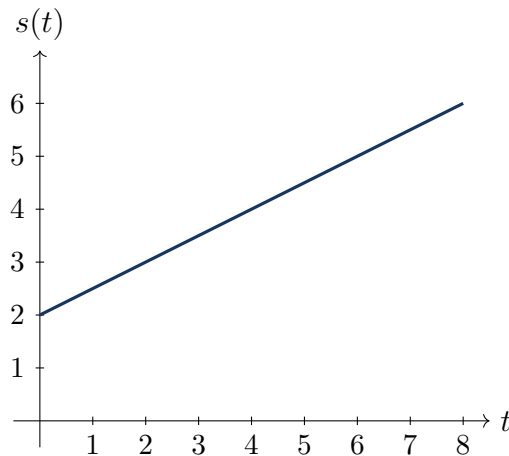
What value do the average velocities appear to approach? $\rule{1.5cm}{0.4pt}$

Part C. Confirm the instantaneous velocity at $t = 1$ using a limit.

Big Idea – Units matter

If position is measured in feet and time in seconds, then velocity is measured in feet per second. In general,

$$\text{units of a derivative} = \frac{\text{units of output}}{\text{units of input}}.$$

Interpreting Velocity from a Graph

If $s(t)$ is a position function, what does this mean about my velocity?

5. Application: Motion on Mars**Example 4: A rock thrown upward on Mars**

A rock is thrown upward on Mars with initial velocity 10 m/s. Its height after t seconds is

$$s(t) = 10t - 1.86t^2.$$

(a) Find the velocity after 1 second.

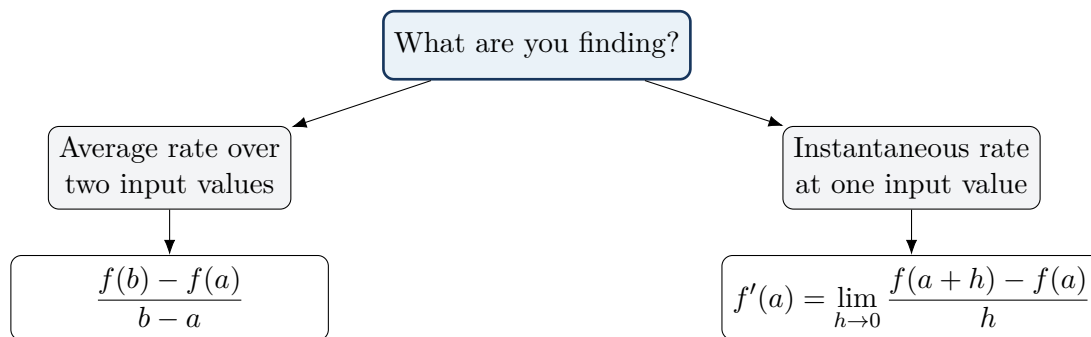
(b) Find the velocity when $t = a$.

Example 4 continued**Interpretation.** What does the sign of velocity tell us?

- $v(t) > 0$: _____.
- $v(t) < 0$: _____.
- $v(t) = 0$: _____.

(c) After how many seconds does the rock hit the surface?At the surface, $s(t) = 0$:**(d) With what velocity does it hit the surface?****Common Mistake**

Do not confuse **height** and **velocity**. To find when the object hits the ground, solve $s(t) = 0$. To find whether it is moving upward or downward, examine the sign of $v(t)$.

6. Choosing the Right Formula

Quick Check

For each question, decide whether you need an average rate of change or a derivative.

1. The average speed of a runner between $t = 2$ and $t = 5$ seconds.
2. The slope of a curve at $x = 3$.
3. The change in population per year from 2020 to 2025.
4. The rate at which temperature is changing at noon.

Reflection

In your own words, how are the following ideas connected?

- average rate of change
- tangent line
- derivative
- velocity

Section 2.7 Summary

- Secant slope: $\frac{f(b) - f(a)}{b - a}$.
- Tangent slope: the limiting value of secant slopes.
- Derivative at a : $f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$.
- Tangent line: $y - f(a) = f'(a)(x - a)$.
- If $s(t)$ is position, then $s'(t)$ is velocity.
- Derivative units are output units per input unit.

Next Time

We will treat the derivative as an entire function, $f'(x)$, rather than a single number $f'(a)$. We'll discover much faster rules for computing derivatives.